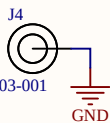


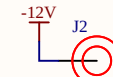
Inverting Buck-Boost



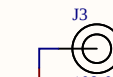
108-0906-001



108-0903-001

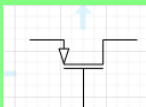


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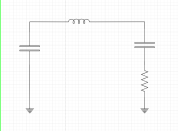
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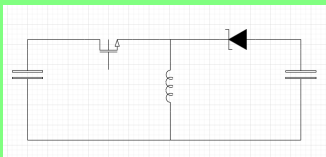


UVLO/OVLO

Input filter



Power stage



PWM

CSP

CSN

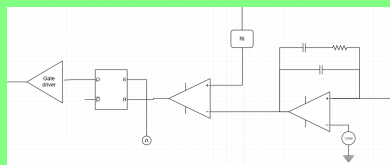
CSP

CSN

Current sense

CS

Control circuit



PWM

OCF

RAMP

FB

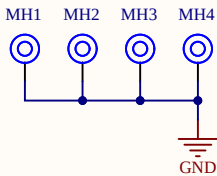
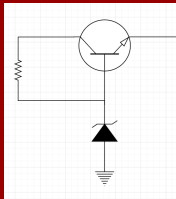
-12V

R1
1.13k
0.1%

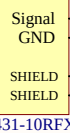
R2
10k
0.1%

GND

Linear regulators

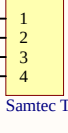


J5



031-5431-10RFX

J6



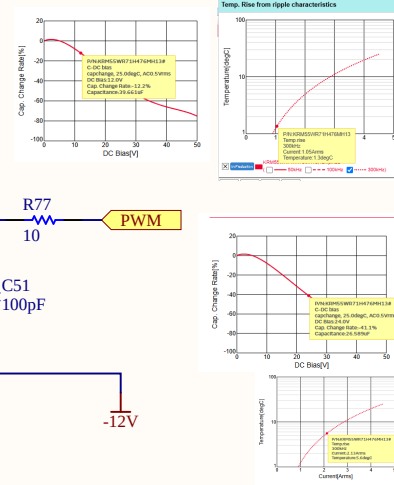
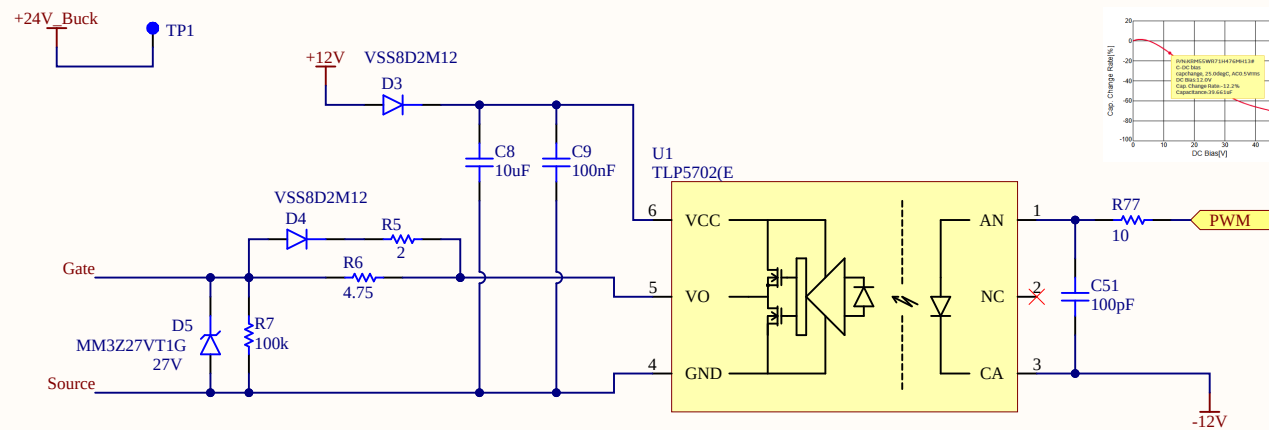
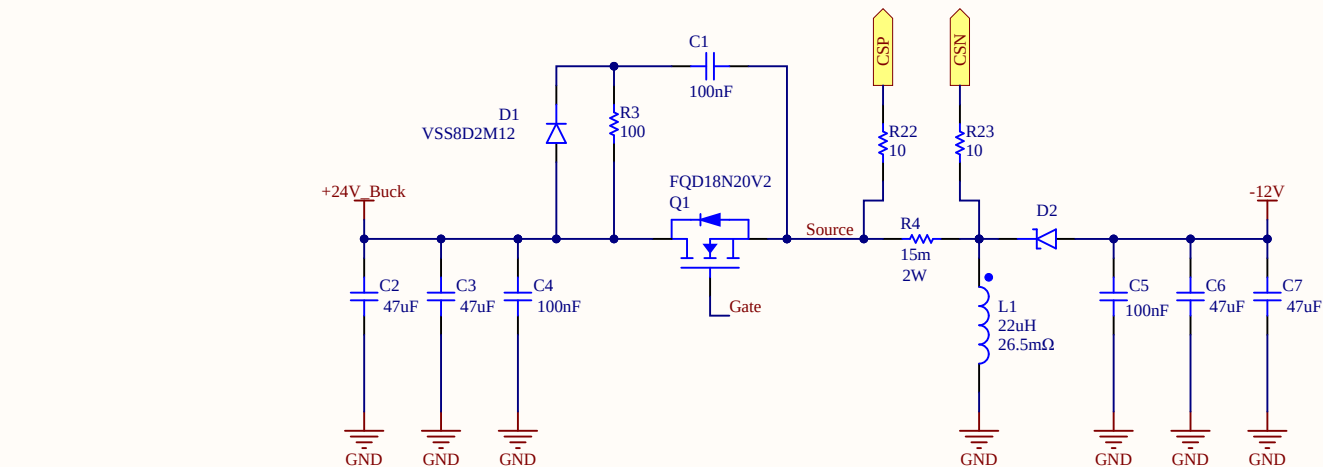
Samtec TS-104-T-AA

J6 is a 4-pin header that selects the op-amp negative input source: either the injected sawtooth signal (to subtract it from the compensator output) or -12 V to disable this feature.

J5 is a female BNC connector used to inject an external compensating ramp (sawtooth) if needed

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Power stage



ANALYSIS

Vin=24V | Vout=-12V Iout=3A | fs=300kHz | D=0.333 | eff=88%

INDUCTOR — SRP1245A-220M (22uH)

IL_avg = Iin + Iout = 1.705 + 3 = 4.705A

$\Delta IL = V_{in} \times D / (f_s \times L) = 1.21A$

IL_peak = IL_avg + $\Delta IL / 2 = 5.31A$

IL_ripple_ratio = 1.21/4.705 = 25.7%

$L_{crit} = (D')^2 \times R / 2f_s = 2.97uH$

CCM confirmed: L=22uH >> Lcrit=2.97uH ✓

MOSFET — FQD18N20V2 (200 V, 15 A, 140 mΩ)

Vds_stress = Vin + Vout = 24+12 = 36V

Vds_rated = 36×1.2 = 43.2V → 600V ✓ (20% derating)

ID_rated ≥ IL_peak = 5.31A → 15A ✓

ID_rms = IL_avg × √D = 2.72A

P_cond = ID_rms² × Rds(on)×1.3 = 2.72²×0.637 = 4.71W (30% derating)

P_sw = 0.5×Vds×IL_peak×(tr+tf)×fs = 0.5×36×5.31×20n×300k = 0.57W

P_total = 4.71 + 0.57 = 5.28W

DIODE — STPS10L60S (60V, 10A, Vf=0.4V, DPAK)

Vr_stress = Vin + Vout = 36V

Vr_rated = 36×1.2 = 43.2V → 60V ✓

ID_avg = Iout = 3A

ID_peak = IL_peak = 5.31A

P_diode = Vf×Iout×(1-D) = 0.4×3×0.667 = 0.8W

OUTPUT CAPACITOR — KRM55WR71H476MH13K ×2 (47uF,50V,X7R)

C_min = Iout×D / (fs×ΔVout) = 27.7uF

C_nominal = 2×47 = 94uF

C_derated @ 12V = 2×39.6 = 79.2uF > 27.7uF ✓

ΔVout_target = 1%×12 = 120mV

ESR_max = 0.5×ΔVout/ΔIL = 49.6mΩ

(the ESR can contribute max by a 50% to the total allowed ripple)

ESR_eff = 4mΩ/2 = 2mΩ << 49.6mΩ ✓

IC_rms = Iout×√(D/D') = 2.12A

IC_rms/cap = 2.12/2 = 1.06A ✓

INPUT CAPACITOR — KRM55WR71H476MH13K ×2 (47uF,50V,X7R)

C_min = Iout×D/(fs×ΔVin×D') = 20.8uF

C_nominal = 2×47 = 94uF

C_derated @ 24V = 2×26.6uF = 53.2uF > 20.8uF ✓

ΔVin_target = 1%×24 = 240mV

ESR_max = 0.5×ΔVin/IL_peak = 22.6mΩ

ICin_rms = IL×√(D×D') = 2.22A ✓

Vrated = 32×1.5 = 48V → 50V ✓

Cin impedance Zcin = 1/(2π×fs×Cin) + ESRcin = 0.413Ω

Cdamp impedance Zcdamp = 1/(2π×fs×Cdamp) + ESRdamp = 0.013Ω

IRMS Total Input Caps (Cin + Cdamp) = IL×√(D×D') = 2.22A

IRMS Cdamp = Zcdamp / (Zcdamp + Zcin) = 2.16A ✓

POWER BUDGET

P_mosfet = 5.28W

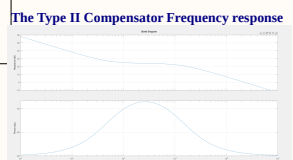
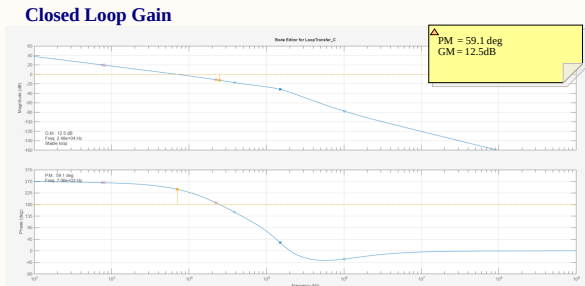
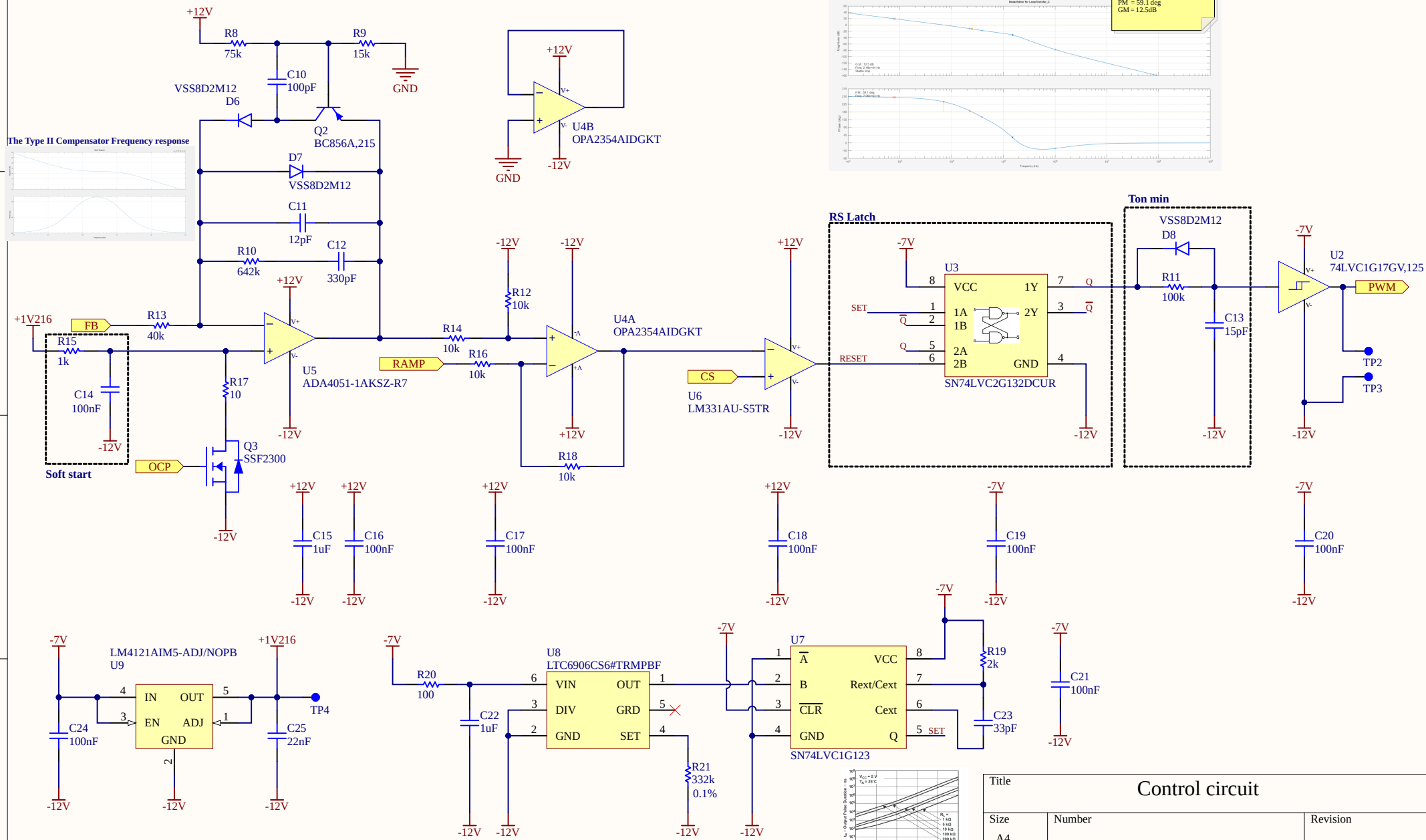
P_diode = 0.80W

P_total = 6.08W

η_estimated = Pout/(Pout+Ploss) = 36/(36+6.08) = 85.6%

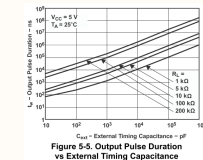
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File:	C:\Users\...\Power stage.SchDoc		Drawn By:	Mohamed Drissi	

Control circuit



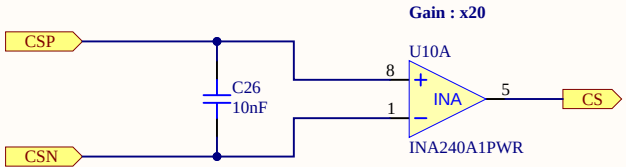
RS Latch

Ton min

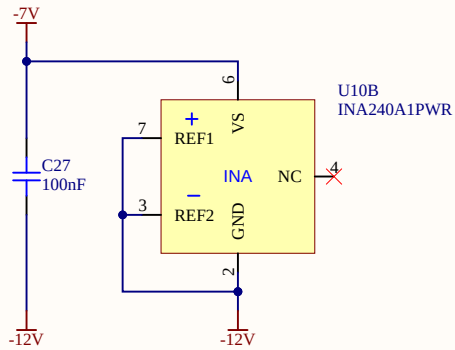


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Control circuit		
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Current sense

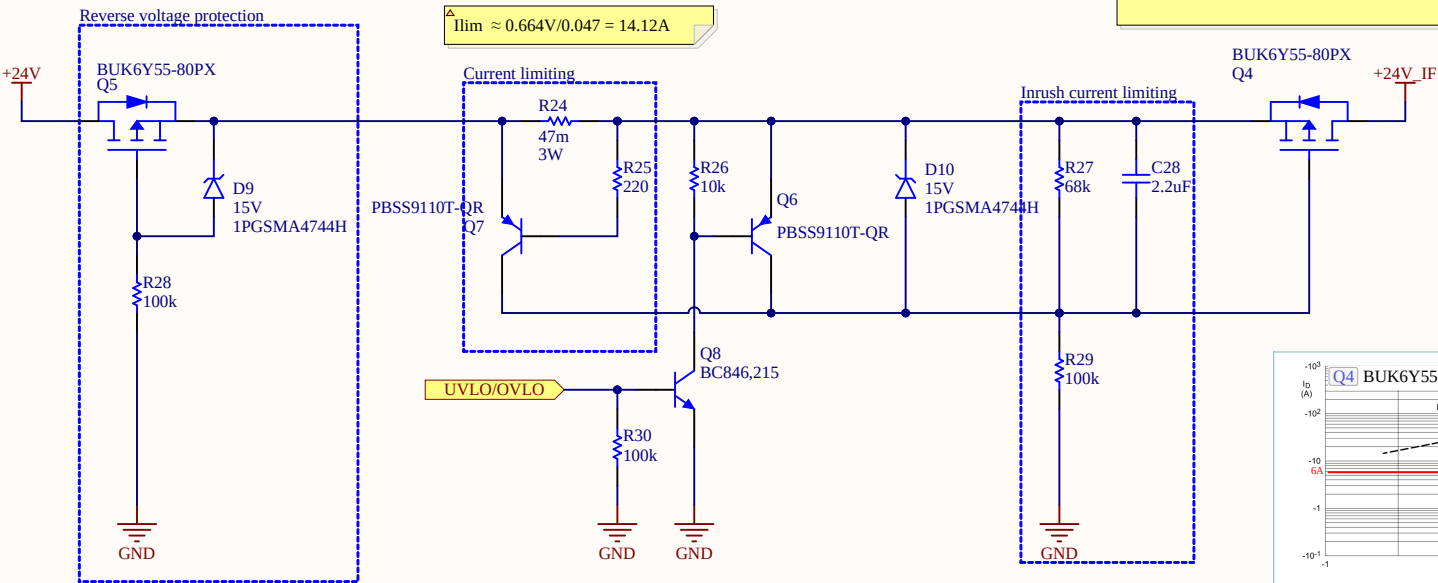


^A Total current gain = 0.015 x 20 = 0.3



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Current sense		
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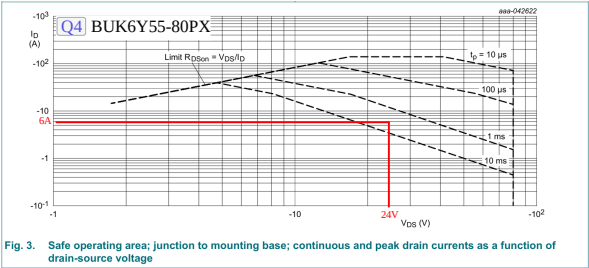
Input switch



△ STATIC POWER DISSIPATION for Q4

$P = I_{in}^2 \times R_{DS(on)}$
 $= (1.705 \text{ A})^2 \times 0.055 \Omega$
 $= 0.160 \text{ W}$

Maximum static power dissipation: 160 mW
Estimated junction-to-case temperature rise: 0.12 °C
(RθJ-MB = 1 °C/W)



△ $C_{load} = C_f + C_{damp} + C_{in}$
 $= 5.45\mu + 53.3\mu + 220\mu = 278.75\mu \approx 300\mu$
 $\Delta t = C_{load} \times V_{in} / I_{inrush_max}$
 $\Delta t = 300\mu F \times 24V / 6A = 1.2ms$ (The MOSFET Q4 can withstand pulses longer than 1 ms. The SOA analysis is a conservative method for verifying whether the transistor can support the inrush current while operating in the linear region. In this analysis, the current is assumed to be a square wave and the VDS is assumed to remain constant at 24 V throughout the entire 1.2 ms inrush period. Based on these conservative assumptions, the MOSFET can be considered suitable for this application.)
based on LTSpice simulation a cap value of 2.2uF C28
in combination with 100K R29 we can achieve approx $I_{inr} = 5.8A$ within 800us which is sufficient and Q4 can withstand
 $V_{gate(steady\ state)} = V_{in} \times R_{29} / (R_{27} + R_{29}) = 24 \times 100k / (68k + 100k) = 24 \times 0.595 = 14.3V$
 $V_{GS,ss} = V_{gate,ss} - V_{in} = 14.3 - 24 = 9.7V$ more than Vgs threshold and less than Vgs max of

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INPUT EMI FILTER : Design Notes

Standard: EN 55032:2015 Class B — Conducted Emissions

Limit @ 300kHz = 56 dBμV (quasi-peak)

Frequency range: 150kHz – 30MHz

SPECIFICATIONS:

Vin = 24V | Vout = -12V | Iout = 3A | fs = 300kHz | D = 0.333 | Cin = 53.2uF (derated @24V)

STEP 1 : NOISE LEVEL ESTIMATE

First harmonic amplitude:

$$I_1 = (2 \times I_L \times \sin(\pi \times D)) / \pi = (2 \times 4.705 \times \sin(\pi \times 0.333)) / \pi = 2.60A$$

Noise voltage across Cin at fs:

$$V_{noise} = I_1 / (2\pi \times fs \times C_{in}) = 2.60 / (2\pi \times 300k \times 53.2u) = 25.9mV = 25.9 \times 10^3 \mu V$$

Estimated noise level:

$$|Noise|_{dB\mu V} = 20 \log(25.9 \times 10^3) = 88.3 \text{ dB}\mu V$$

Required attenuation:

$$Att = |Noise|_{dB\mu V} - V_{max} = 88.3 - 56 = 32.3 \text{ dB}$$

Adding 10dB safety margin:

$$Att_{target} = 32.3 + 10 = 42.3 \text{ dB}$$

STEP 2 : INDUCTOR SELECTION

Lf = 10uH → Selected: 10uH, 12A, SMD

STEP 3 : CAPACITOR SELECTION

Resonance one decade below fs:

$$C_f = C_{in} / (C_{in} \times L_f \times (2\pi \times fs / 10)^2 - 1) = 53.2u / (53.2u \times 10u \times (2\pi \times 30k)^2 - 1) = 3\mu F$$

Required attenuation:

$$C_f = 1 / L_f \times (10 \wedge (Att/40) / (2\pi \times fs))^2 = 5\mu F$$

Select higher value: Cf = max(5μF, 3μF) = 5μF → 10μF selected derated to 5.45μF @24V

NOTE: Cf is a SEPARATE dedicated filter capacitor distinct from Cin.

$$f_{c_filter} = 1 / (2\pi \times \sqrt{L_f \times C_f}) = 1 / (2\pi \times \sqrt{10u \times 5.45\mu F}) = 21.6kHz$$

Attenuation verification:

$$A = 40 \times \log(fs / f_{c_filter}) = 40 \times \log(300k / 21.6k) = 45.7dB \gg 42.3dB \checkmark$$

STEP 4 : DAMPING CAPACITOR

$$C_d \geq 4 \times C_{in} = 4 \times 53.2u = 212.8uF \rightarrow 220uF \text{ selected}$$

Required ESRd:

$$ESR_d \leq \sqrt{L_f / C_{in}} = \sqrt{10u / 53.2u} = 0.434\Omega$$

Damping resistor power:

$$P_{R_damp} \approx 0.5 \times \Delta I_{in}^2 \times R_{damp} = 0.5 \times (1.21)^2 \times 1.5 = 0.8W$$

Selected Cd: 220uF 35V polymer electrolytic ESR_polymer ≈ 11mΩ < 434mΩ ✓

NOTE: Polymer ESR too low, added series resistor Rd = 0.4Ω in series with Cd to meet ESRd target

Voltage rating: Vin_max × 1.25 = 32 × 1.25 = 40V → 50V ✓

$$C_{in} \text{ impedance } Z_{cin} = 1 / (2\pi \times fs \times C_{in}) + ESR_{cin} = 0.413\Omega$$

$$C_{damp} \text{ impedance } Z_{cdamp} = 1 / (2\pi \times fs \times C_{damp}) + ESR_{damp} = 0.013\Omega$$

$$IRMS \text{ Total Input Caps } (C_{in} + C_{damp}) = I_L \times \sqrt{D \times D'} = 2.22A$$

$$IRMS \text{ Cdamp (current divider)} = Z_{cin} / (Z_{cdamp} + Z_{cin}) = 0.062A \checkmark$$

STEP 4 : MIDDLEBROOK CRITERION

$$Z_{in_conv} = V_{in} \times \eta / P_{out} = 576 \times 0.88 / 36 = 14.1\Omega$$

Full frequency spectrum verified using matlab: see Plot

EMC STANDARD : EN 55032:2015

EN 55032 "Electromagnetic Compatibility of Multimedia

Equipment". It defines conducted emission limits on the power port for Class B equipment (residential environment):

150kHz – 500kHz : 56–66 dBμV (quasi-peak)

46–56 dBμV (average)

500kHz – 30MHz : 56 dBμV (quasi-peak)

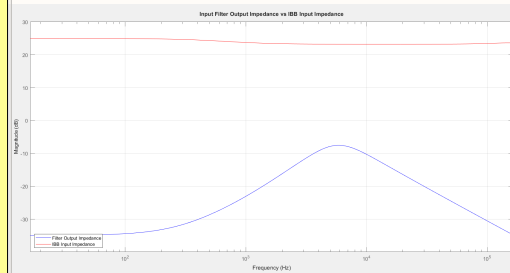
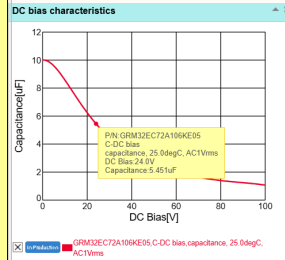
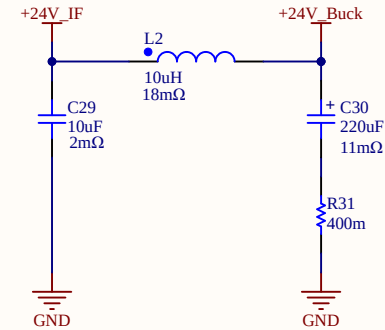
46 dBμV (average)

Worst case for this design at fs=300kHz: 56 dBμV QP.

Reference : Conducted EMI Characteristics And Mitigation Technique

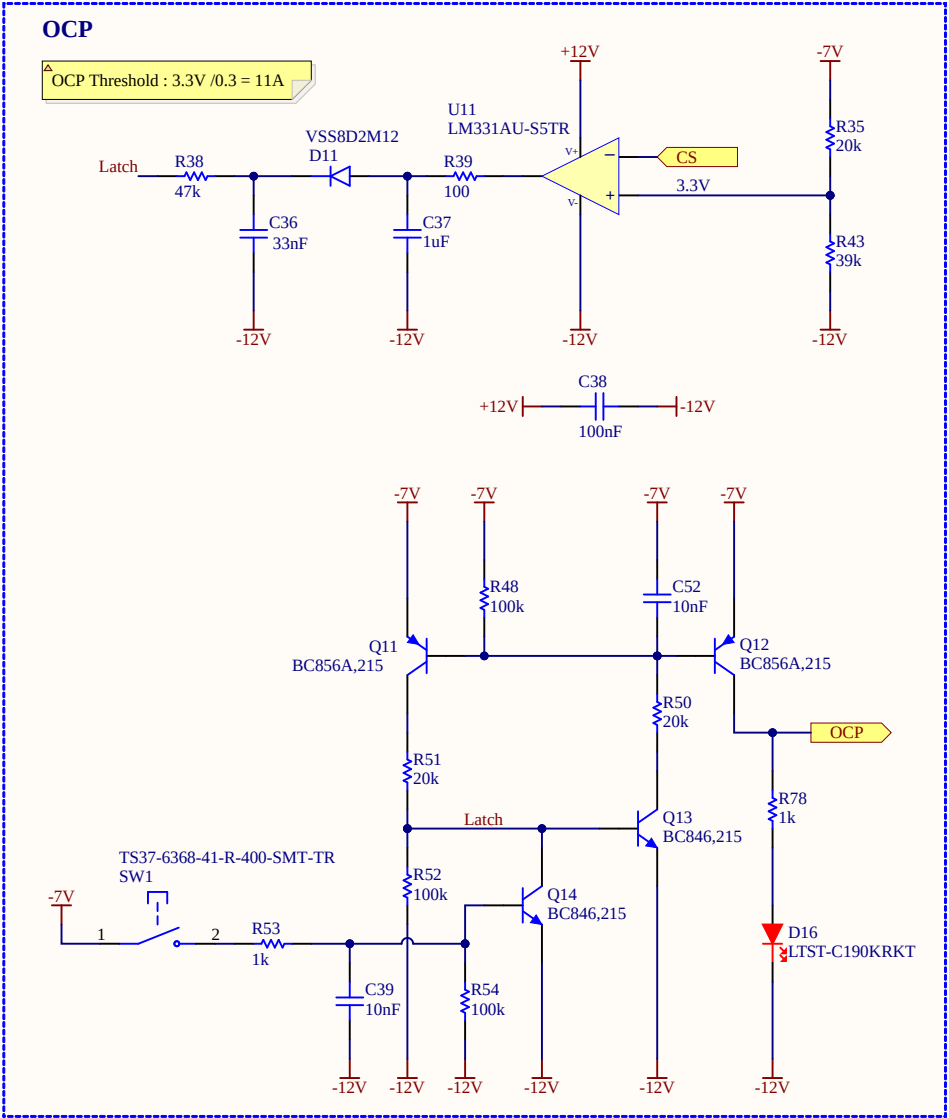
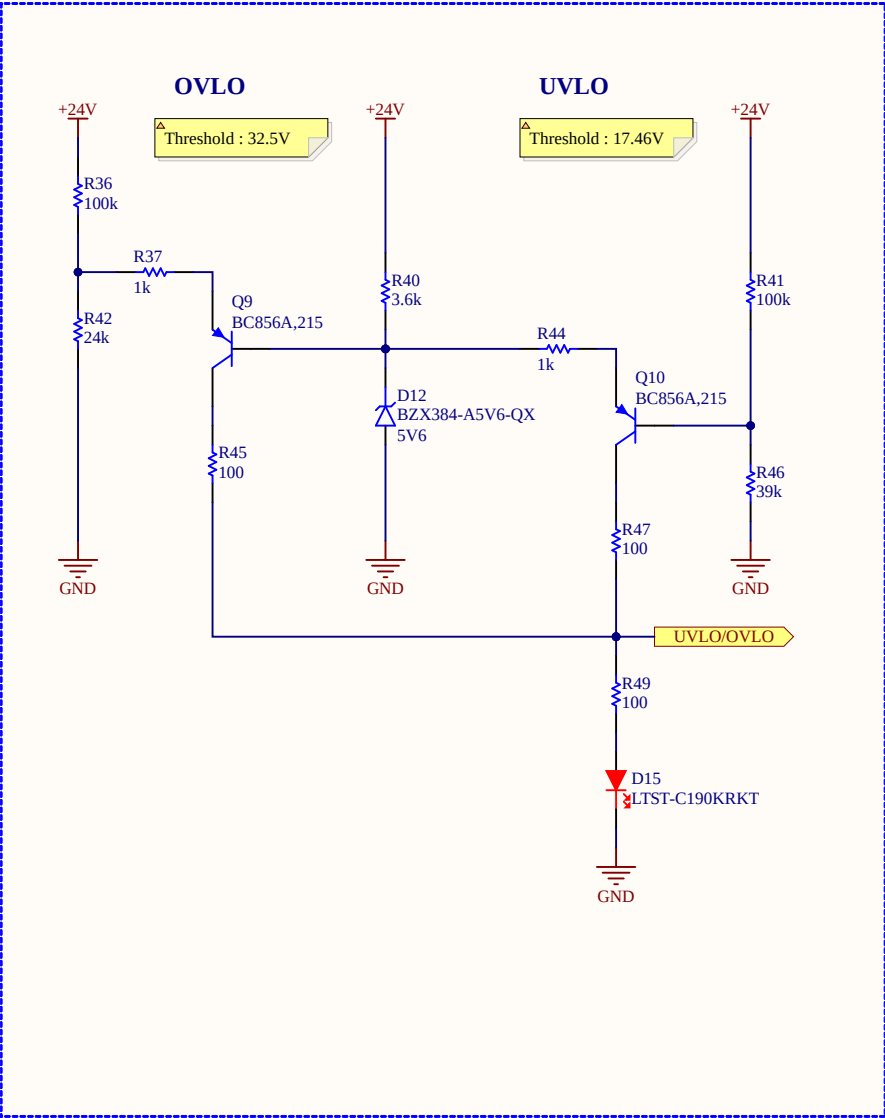
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Input filter



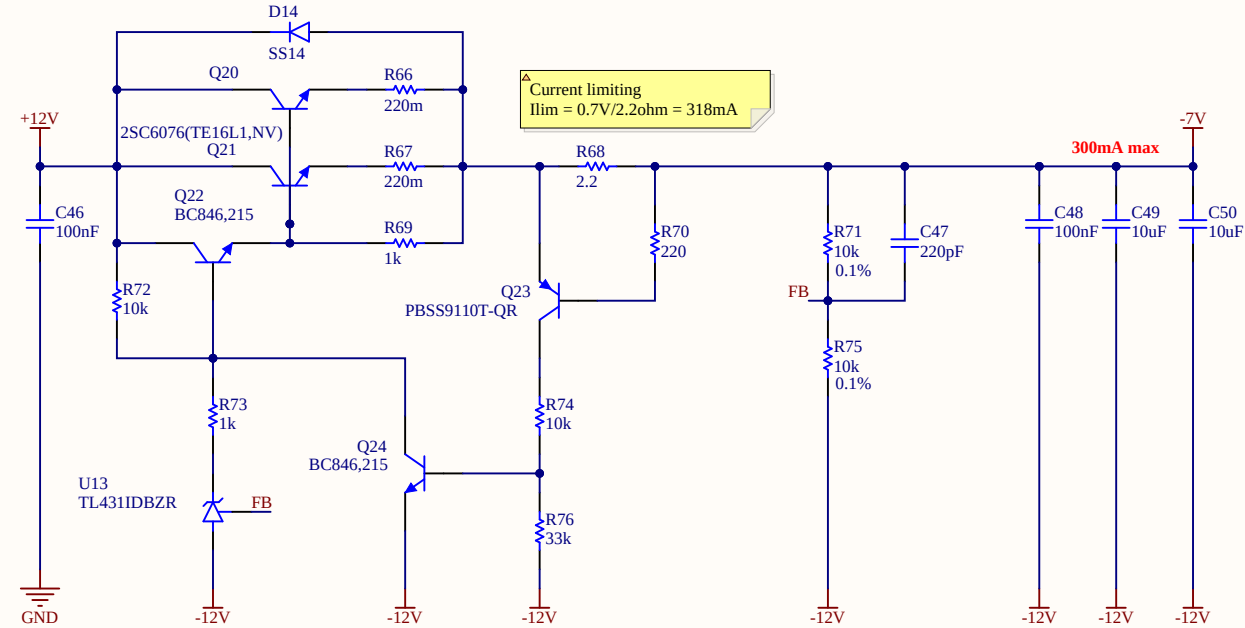
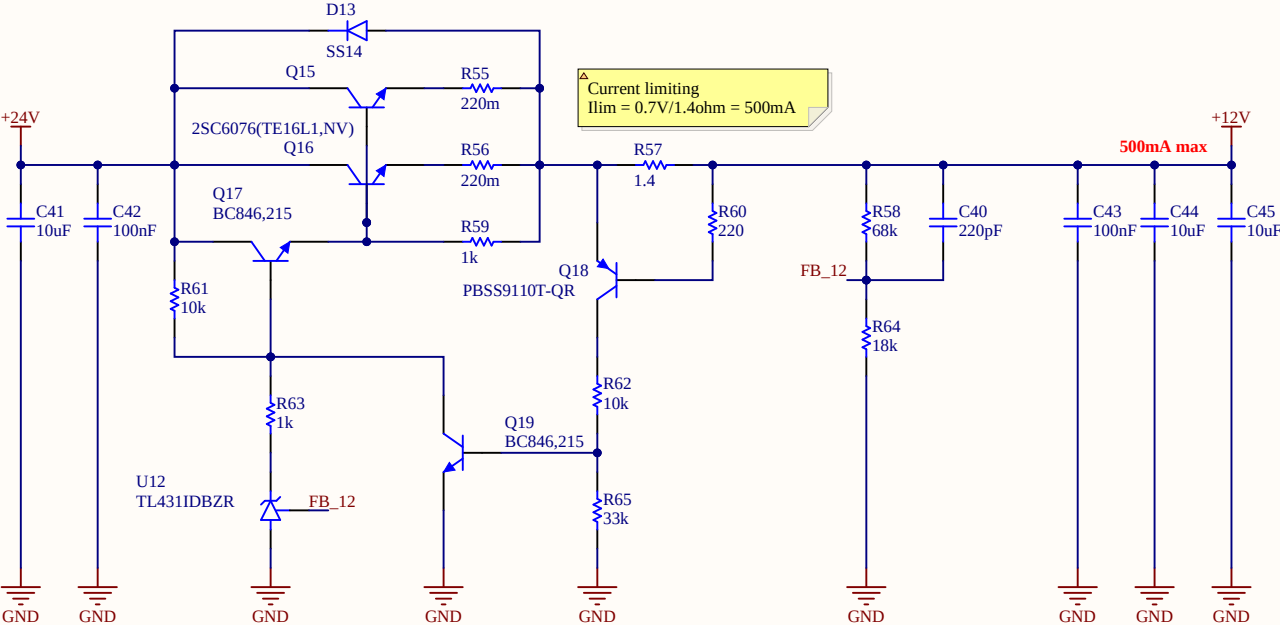
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Protections

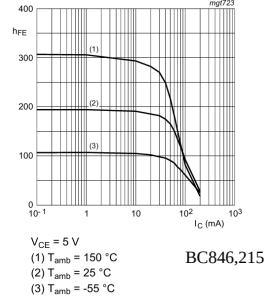
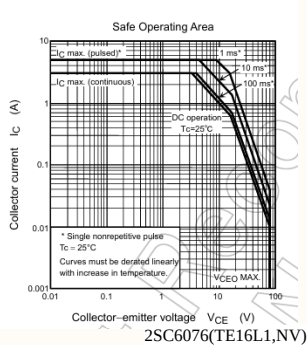


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Linear regulators



To obtain the maximum possible current gain using Darlington pair it's important to ensure that the driver transistor has enough collector current to ensure acceptable hFE. When run at very low collector current, most transistors will have a gain that's well below the quoted figure in the datasheet. The base-emitter resistor (shown in the 12V regulator) is sized to ensure that the drive transistor has a collector current that's above the minimum. The resistor also helps to ensure a faster turn-off and minimises leakage current.



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Linear regulators		
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